

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

low-pass filter at $\omega=1.0$ for different rectangular window lengths (N) of 49, 99 and 149.

We will see frequency responses generated by the software for high-pass, band-pass and band-reject filters with various values of N and ω .

Frequency response of high-pass filter with $N=61$ and $\omega=1.2$ is shown in Fig. 3. Frequency response of band-pass filter with $N=61$, $\omega_1=1.2$ and $\omega_2=2.1$ is shown in Fig. 4.

Frequency response of band-reject filter with $N=91$, $\omega_1=1.2$ and $\omega_2=2.1$ is shown in Fig. 5.

Further, it may be observed that when ' N ' increases, the width of the pass-band decreases and the slope of the transition band increases. The number of ripples increases in the

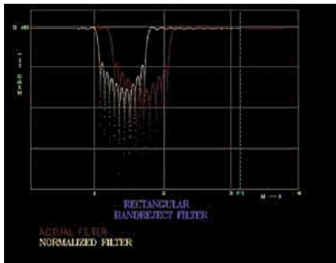


Fig. 5: Frequency response of band-reject filter with $N=91$, $\omega_1=1.2$ and $\omega_2=2.1$

pass-band with the amplitude of the end-ripple remaining the same, which however gets shifted towards the edge and finally out of the pass-band for larger values of ' N '. The width of the lobes decreases in the band-reject and the band-reject attenuation also increases with the increase in ' N ' as more lobes enter the transition band.

Moreover, the relative amplitude of the lobes remains constant.

Future applications

These filters can be used in a wide variety of applications like noise removal, selective passing of frequency contents in voice or any real-time waveforms. Using these filters' sub-routine, you can design your own software for DSP applications or use this code to be translated in a hardware description language like Verilog or VHDL so it can be burnt into FPGA. ●

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